ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER: 2021-2022**
DURATION: 3 Hours **FULL MARKS: 150**

CSE 4303: Data Structures

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all** questions. Marks of each question and corresponding CO and PO are written in the right margin.

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1. a) Suppose that you have a *reference* to a node in a *singly linked list* and this node is not the last one in the list. — 5+5, CO2, PO2
   i. How could you *remove* the element at this node from the list, while maintaining a proper linked list data structure? Note that this would reduce the number of nodes by 1. The solution should not require looking for this node by starting at the head of the list, rather it must be done in constant time i.e., $O(1)$. (The solution is just a few lines of code).
   ii. How could you insert an element into the list at the position that comes *before* the element at this given node?

   b) Consider the polynomial: — 4+4, CO2, PO1

$$5y^2 - 3y + 2$$

   Represent the polynomial as an '*expression tree*' where the internal nodes can only contain the operators. Utilize the newly generate tree to convert the polynomial as *postfix* expression.

   c) Illustrate the process to maintaining a *Queue* using two *stacks*. — 7, CO2, PO3

2. a) Suppose a *Hash function* is defined as '$hash(x) = x \ mod \ 10$'. A set of number is given as $(4371, 1323, 6173, 4199, 4344, 9679, 1989)$. Insert the numbers inside a hash table using the following collision resolution strategies: — 5×3, CO1, PO1
   i. Separate chaining
   ii. Quadratic probing
   iii. Double hashing; considering a second hash function $f(i) = i \times hash_2(x)$ and $hash_2(x) = 7 - (x \ mod \ 7)$.

   b) Design an algorithm to implement a *First-In-First-Out Queue* using a *Priority Queue.* — 10, CO3, PO3

3. a) Canadian postal codes are of the form '$L_1D_1L_2D_2L_3D_3$', where 'L' is always a letter (A-Z) and 'D' is always a digit (0-9). Suppose you have a company and you wish to keep track of customers who have similar postal codes, so the map might be (postal code, list of customers). Let the letters A-Z be coded with numbers 1 to 26, for example $code(B) = 2$, and let the digits be coded by their numerical values. — 3+5, CO3, PO2
   i. For a hash function:

$$hash(L_1D_1L_2D_2L_3D_3) = \left(\sum_{i=1}^{3} code(L_i) + \sum_{i=1}^{3} code(D_i)\right) mod \ 10$$

   Give an example of a postal code that begins with $H3A$ and collides with $H2B5X7$.
   ii. Give an example of a *hash function* that would never result in a collision. How large would the hash table need to be?

   b) Both *hashing* and *binary search trees* allow to efficiently store and access map entries. What are the advantages/disadvantages of each, which would make you choose one over another? — 5, CO3, PO2

   c) Given a set of numbers stored in a *Priority Queue*, design an algorithm to add an amount 'x' to the ith location. Here 'x' is an integer value and 'i' represents the index (starting from 1). Assume that the number with the higher value is considered to have the lower priority. — 12, CO2, PO3

4. a) Draw all *binary search trees* of height 2 that can be made from *all* the letters ABCDEF, assuming their natural ordering. 5 CO1 PO2

b) Form a *binary search tree* of strings from the sentence: 3+2 CO1 PO1

"$Form\ a\ binary\ search\ tree\ of\ strings$"

In particular, the tree must be constructed by adding the words in the order they are found in the sentence, namely first add the word "Form" to an empty tree, then add the word "a", etc. Mention the status of the tree after removing the word "Form".

c) Two *binary trees* A and B are said to be *isomorphic* if, by swapping the left and right subtrees of some nodes of A, one can obtain a tree identical to B. For example, the following two binary trees are isomorphic. 10+5 CO2 PO3

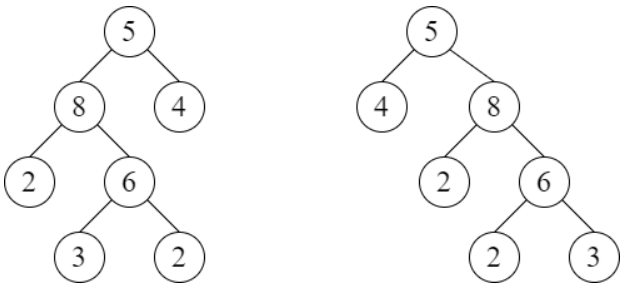


Figure 1: Binary Tree for Question 4(a)

i. Design a recursive algorithm $isIsomorphic(n1, n2)$ for checking if two binary trees are isomorphic. The arguments should start with references to the root nodes of the two trees.

ii. Simulate the proposed algorithm using the trees mentioned in Figure 1 and show that the algorithm is functioning properly.

5. a) *FIFA World Cup 2022* is going on, and as expected, thousands of '*pages*' (with unique names) have been created inside '*Facebook*'. Facebook is providing two types of search criteria so that the fans can easily search their desired pages. If the '*query*' directly matches with the title of any existing page, it only shows that in suggestion. Otherwise, it shows all the pages that start with that query string. Discuss an effective approach to satisfy the requirements. (Pseudocodes not required) 7 CO3 PO2

b) $7, 5, 4, 21, 2, 10, 14, 8, 1, 9$ 3×6 CO2 PO1

Showing detailed steps, perform the following operations on a *Segment Tree* (Array index starts from 1):

i. Build the tree using the above-mentioned numbers where the internal nodes represent the sum of values within a given range.

ii. Increment all values within the range (4,8) by 5.

iii. Increment all values within the range (1,9) by 10.

iv. Show the summation of the range (2,4).

v. Show the summation of the range (3,7).

vi. Add 7 to the value at index-8.

6. a) Define *Data Structure*. Mention separate use-cases for *eight* different data structures. 2+ 1.5×8 CO1 PO1

b) Consider the following sequence of *stack* commands: 5 CO1 PO1

$push(a), push(b), push(c), pop(), push(d), push(e), pop(), pop(), pop(), pop()$

i. What is the order in which the elements are popped?

ii. Considering the positions of the $push()$ commands to be fixed, change the position of the $pop()$ commands in the so that the items are popped in the order: 'b, d, c, a, e'.

c) How is constant time accessing ensured for an array of integers, whereas, not for Linked Lists? Does this property contradict in case of an array of strings, in which the strings have possibly different lengths? 4+2 CO1 PO2

